
When Does a Math Model Know the Reasoning Went Wrong?

Tracing the First Error in Hidden States

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Abstract

1 We ask whether a math-specialized model’s hidden states register the first invalid
2 step. At every retained step boundary in the 3,400-trace ProcessBench dataset, we
3 record the residual-stream state of Qwen2.5-Math-1.5B-Instruct and train layer-
4 wise linear probes to predict whether the solution is invalid so far. With problem-
5 grouped splits and validation-only model selection, the selected probe obtains
6 held-out AUROC 0.866 (95% whole-trace bootstrap CI [0.849, 0.884]), compared
7 with 0.733 for current-step TF-IDF and 0.730 for position. The same score
8 reaches AUROC 0.873 when restricted to erroneous traces, and source-specific
9 directions transfer across every pair of four mathematical domains (mean off-
10 diagonal AUROC 0.805). Temporal localization is harder: the probe identifies
11 the exact first error on 28.9% of erroneous traces, lands within one step on 59.3%,
12 and is systematically late by 0.59 steps among detections. Finally, adding or sub-
13 tracting the probe direction at the boundary produces no detected change in the
14 model’s explicit correctness verdict. Because that native verdict readout is itself
15 degenerate (AUROC 0.283), the intervention is inconclusive rather than evidence
16 of causal non-use. The hidden states contain a transferable invalidity signal, but
17 the experiment does not show precise timing or behavioral use.

18 1 Introduction

19 A solution can reach a wrong answer through one early mistake, or reach the right answer through
20 an invalid derivation. Outcome accuracy hides this distinction, and verifiers increasingly score in-
21 termediate steps rather than only final answers (Cobbe et al., 2021; Lightman et al., 2023). Yet an
22 external verifier leaves open a mechanistic question: when a solution first becomes invalid, does the
23 solving model’s own representation change in a detectable way?

24 An internal signal near the first invalid step would show that the model tracks errors during an
25 extended computation, and it might support lightweight detection. Three claims have to be kept apart.
26 A state may contain decodable information without localizing the transition; a direction may rank
27 examples across domains without transferring its threshold; and the model’s output computation
28 need not use a decodable direction (Hewitt and Liang, 2019; Elazar et al., 2021).

29 We study these distinctions in Qwen2.5-Math-1.5B-Instruct (Yang et al., 2024) on ProcessBench,
30 whose human annotations identify the first erroneous step in solutions drawn from GSM8K, MATH,
31 OlympiadBench, and Omni-MATH (Zheng et al., 2025). Our experiment was specified before test
32 results were inspected. We extract one residual-stream state per layer and written step, fit regularized
33 linear probes on problem-grouped splits, evaluate first-threshold-crossing localization, test source-
34 to-source transfer, and intervene along the learned direction.

35 Our main findings are:

- Invalid-so-far is linearly decodable at the selected hidden-state index with AUROC 0.866, exceeding position, current-step lexical, and shuffled-label controls. Discrimination remains high within erroneous traces (AUROC 0.873).
- The ranking signal transfers across all 12 off-diagonal source pairs, but exact localization is weaker and less calibrated: only 28.9% of erroneous traces are localized exactly.
- The preregistered additive intervention yields no signed dose response. However, the underlying explicit verdict score fails to distinguish valid from invalid boundaries, so this result cannot say whether the decoded signal is causally used.

2 Related work

Process-level mathematical verification. Sampling solutions and ranking them with a verifier improved GSM8K performance before modern process supervision (Cobbe et al., 2021). PRM800K then enabled direct comparison of outcome and human step-level supervision; process supervision performed better on a representative MATH subset (Lightman et al., 2023). Later step-aware verifiers and process reward models used intermediate correctness for search and training (Li et al., 2023b). These reward models can generalize poorly from GSM8K and MATH to harder competition problems. ProcessBench addresses this problem by annotating the earliest error in 3,400 traces spanning four sources and evaluating both reward and prompted critic models (Zheng et al., 2025). Prompted LLM studies likewise find that identifying and correcting mathematical errors is a separate capability from solving, and that exact first-error localization remains difficult even when a reference solution is provided (Li et al., 2024; Srivatsa et al., 2025).

We use ProcessBench differently from its standard benchmark setting. We repartition it to analyze one model’s representations, so our metrics are not leaderboard scores. Rather than score generic step quality, we ask whether a prefix has crossed the human-annotated first-error boundary and when the probe first detects that change.

Internal correctness and reasoning signals. Models can sometimes estimate whether their own answers are correct (Kadavath et al., 2022), and truth-related information can be recovered from activations even when outputs are unreliable (Burns et al., 2023). Work closest to ours shows that lightweight probes detect arithmetic errors and can guide selective re-prompting (Sun et al., 2025); ReProbe uses learned hidden-state scorers for multi-domain step verification (Ni et al., 2026); and STEP uses hidden states to prune low-quality reasoning traces during test-time scaling (Liang et al., 2026). Circuit analysis of simple arithmetic further finds a dissociation between computation and validation, with smaller models relying heavily on surface-level numerical consistency heads (Bertolazzi et al., 2025).

These studies show that internal states can predict reasoning quality. We use human first-error labels for open-ended model-generated solutions from grade-school through Olympiad mathematics. We measure the first score crossing rather than only step classification, test every source-to-source transfer, and evaluate the probe direction with a held-out causal intervention.

From probing to causal evidence. Probe accuracy alone is ambiguous because a classifier may exploit accessible but behaviorally unused information; controls and constrained probe families help but do not establish use (Hewitt and Liang, 2019; Belinkov, 2022). Causal work instead modifies representations through amnesic removal (Elazar et al., 2021), causal tracing (Meng et al., 2022), activation addition, or inference-time steering (Turner et al., 2023; Li et al., 2023a). We therefore treat intervention as a stronger, separate test. A null intervention is interpretable only when its output assay is sensitive and meaningful. Our original verdict prompt fails that condition.

3 Method

3.1 Data, model, and leakage-resistant splits

ProcessBench contains 3,400 model-generated solutions with a human label e_i : the zero-indexed first erroneous step, or -1 if every step is correct (Zheng et al., 2025). Its four sources span grade-school word problems, competition mathematics, and Olympiad problems (Cobbe et al.,

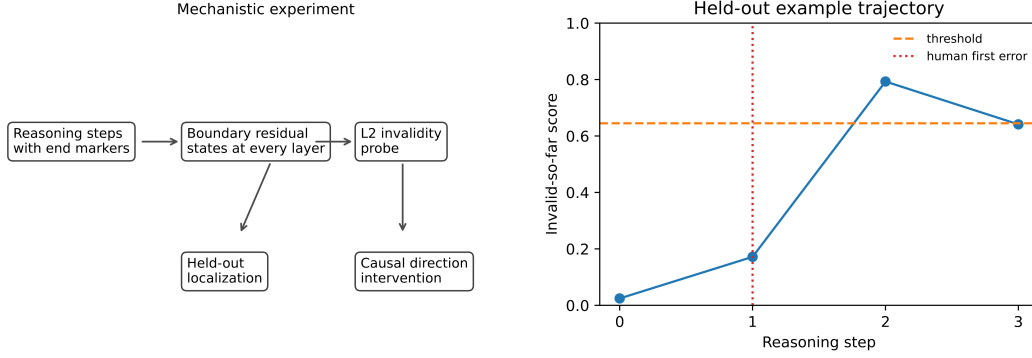


Figure 1: Method and one held-out score trajectory. We probe boundary residual states at every layer, select all hyperparameters on validation data, and define the predicted first error as the first threshold crossing. In the example, the human error begins at step 1 and the score crosses at step 2. High step AUROC therefore need not imply exact localization.

2021; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024). We evaluate the 1.5B-parameter instruction-tuned Qwen2.5-Math model in bfloat16 with a 2,048-token context limit; over-length traces are excluded rather than truncated.

All traces sharing whitespace-normalized, case-folded problem text form one group. Within source $\times$ error-status strata, a deterministic seed-42 allocation places approximately 60%/20%/20% of groups in train/validation/test. All steps from a problem therefore remain in one partition. The published test artifact contains 669 traces (432 erroneous and 237 fully correct) and 4,985 step boundaries.

3.2 Boundary representations and targets

The prompt renders each step k followed by a unique END_STEP marker. One causal forward pass records the residual-stream state $h_{ik}^{(\ell)} \in \mathbb{R}^{1536}$ at each marker across 29 hidden-state indices. Although later steps occur in the same prompt, causal attention prevents them from influencing an earlier boundary state. For trace i and step k , the primary target is

$$y_{ik} = \mathbb{I}[e_i \geq 0 \wedge k \geq e_i], \quad (1)$$

which denotes that the displayed reasoning is *invalid so far*. It is intentionally persistent after onset and should not be read as a label of the current step in isolation.

3.3 Predictive, localization, and transfer tests

At each index ℓ , we standardize dimensions using training data and fit a class-balanced, L2-regularized logistic probe. Validation AUROC selects $C \in \{0.01, 0.1, 1, 10\}$; validation Process F1 selects a threshold from 0.05 to 0.95; and validation Process F1 (then AUROC) selects the layer. The chosen probe is refit on train+validation and evaluated once on test. The selected setting is index 23, $C = 0.01$, threshold 0.645.

For a trace, the predicted first error is the first score at or above this threshold, or -1 if none crosses. We report error-trace exact accuracy, correct-trace rejection accuracy, their harmonic mean (Process F1), complete-trace exact outcome, and tolerance-based localization. We also report step AUROC within erroneous traces, preventing success from a constant distinction between wholly correct and erroneous solutions. Controls use absolute/fractional position, current-step TF-IDF unigrams and bigrams, and shuffled training labels. Uncertainty uses 1,000 percentile bootstrap resamples of whole traces.

For transfer, a separate index-23 probe is trained on each source and evaluated on every test source. Its threshold is fixed using only the training source’s validation set. AUROC therefore tests ranking transfer, whereas Process F1 additionally tests probability and threshold transfer.

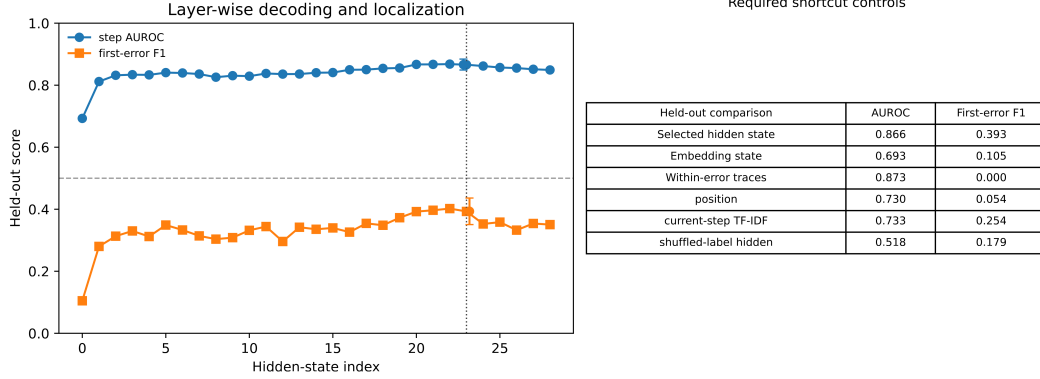


Figure 2: Held-out decoding, localization, and controls. Error bars at the selected index are 95% whole-trace bootstrap intervals. The hidden-state probe ranks invalid prefixes well, but its first-error Process F1 is much lower than its step AUROC. “Within-error traces” has no meaningful correct-trace rejection term, so its Process F1 is shown as zero.

3.4 Probe-direction intervention

The standardized probe weight is transformed back to raw activation coordinates and normalized to v . At the input to decoder layer 23, we replace one held-out boundary activation with

$$h' = h + \alpha \sigma_v v, \quad \alpha \in \{-4, -2, -1, 0, 1, 2, 4\}, \quad (2)$$

where σ_v is the train+validation standard deviation of projections onto v . We sample one boundary from each of 256 test traces, balanced between valid-so-far and invalid-so-far. The outcome is the difference between length-normalized teacher-forced log probabilities of INCORRECT and CORRECT. Twenty matched random unit directions orthogonal to v serve as controls at $\alpha = \pm 4$.

4 Results

4.1 Invalidity is decodable, but first-error timing is harder

Figure 2 shows a rapid rise in decoding performance after the embedding state and stable AUROC above 0.82 through most layers. The validation-selected index obtains test AUROC 0.866 (95% CI [0.849, 0.884]), average precision 0.843 ([0.813, 0.870]), and step F1 0.743 ([0.716, 0.769]). Restricted to erroneous traces, it distinguishes pre-error from error-and-later steps with AUROC 0.873 and average precision 0.925.

The hidden-state score exceeds current-step TF-IDF (AUROC 0.733, Process F1 0.254), position (0.730, 0.054), and the shuffled-label hidden-state control (0.518, 0.179). These gaps argue against simple position and current-step lexical explanations but do not exclude them; the present artifacts lack paired confidence intervals for probe-minus-control differences.

Exact timing is substantially weaker. Across erroneous traces, 28.9% are localized exactly, 59.3% within one step, and 72.7% within two. The detector crosses somewhere on 88.7% of erroneous traces but falsely crosses on 38.8% of fully correct traces. Complete-trace exact accuracy is 0.404; Process F1 is 0.393 (95% CI [0.351, 0.436]). Among detected erroneous traces, the mean predicted-minus-human error is +0.59 steps (95% CI [+0.36, +0.80]): the signal is systematically late rather than a sharp internal alarm at the annotated boundary.

4.2 Ranking transfers better than localization thresholds

Every off-diagonal transfer AUROC in Figure 3 exceeds 0.739. Mean off-diagonal AUROC is 0.805, compared with 0.845 on the diagonal. The sources therefore share part of the ranking signal. Thresholded localization is less stable: mean off-diagonal Process F1 is 0.261 versus 0.336 on the diagonal, with individual transfers ranging from 0.055 to 0.371. Thus a source-trained direction often

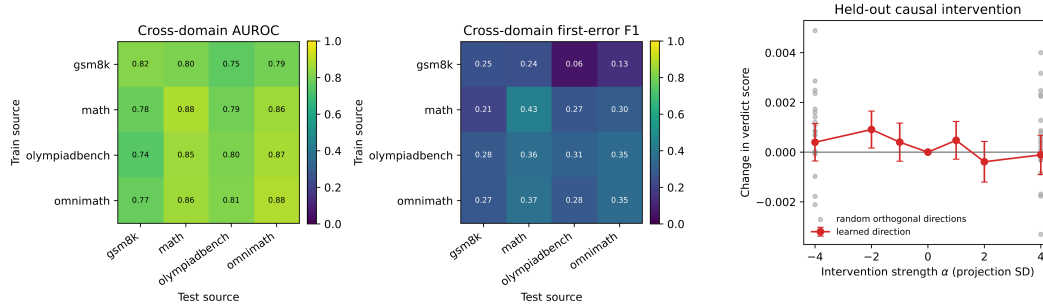


Figure 3: Cross-source transfer and causal intervention. Rows are probe-training sources and columns are held-out test sources. Ranking transfers consistently, while threshold-dependent first-error F1 does not. Right: mean paired verdict changes with 95% whole-trace bootstrap intervals; gray points are matched random orthogonal directions at the extreme doses. No learned dose differs reliably from zero.

orders target-domain boundaries well without transferring calibrated probabilities or the optimal first-crossing threshold.

4.3 The causal assay is inconclusive

Equation 2 produces neither the predicted signed monotonic response nor a nonzero effect at any dose. The mean changes at $\alpha = -4$ and $+4$ are $+0.00040$ ($[-0.00108, 0.00185]$) and -0.00011 ($[-0.00169, 0.00138]$), respectively. Neither extreme beats the distribution of 20 random orthogonal directions.

This null cannot support causal non-use because the unmodified output assay is not informative. All 256 baseline margins favor INCORRECT; recall is 1, specificity is 0, and AUROC is 0.283. Valid boundaries even have a larger mean incorrect-minus-correct margin (5.860) than invalid boundaries (5.625). We can conclude only that this additive intervention, at index 23 and the marked boundary, did not detectably affect this degenerate teacher-forced verdict score.

5 Discussion and limitations

The probe separates prefixes around human-annotated errors and transfers in ranking, but it often fires late and is not shown to drive a correctness judgment. A post-hoc analysis of the frozen test scores found 28.9% exact localization versus 16.4% after circular within-trace shifts ($p = 0.0002$). The score rose by 0.257 at the annotated onset; matched transitions in correct traces rose by 0.113, a difference of 0.144 (95% CI $[0.096, 0.193]$). First-step centering left pooled within-error-trace AUROC at 0.881. Two further post-hoc tests refine this picture. A probe trained on matched boundary-to-boundary transitions distinguishes error onsets from metadata-matched correct-trace transitions at AUROC 0.769 (CI $[0.713, 0.820]$), above the position (0.63), TF-IDF (0.67), and shuffled (0.58) controls, so a representation change at the onset itself is linearly decodable. Re-probing the natural last token of each step instead of the artificial marker reproduces the pipeline at AUROC 0.868 versus 0.866 (paired difference $+0.002$, CI $[-0.007, +0.011]$), so the signal does not depend on the step marker. These checks support a gradual temporal change, not a precise alarm, and remain post-hoc.

The score may still encode generator identity, trace-wide difficulty, final-answer outcome, or features beyond TF-IDF. The persistent target may reward accumulated corruption or create a summary state; the marker-dependence concern is addressed by the boundary control above, but the other shortcut channels remain untested. We test one model and split at one token location, layer, and intervention. The compact release lacks retained train/validation and exclusion counts. Paired bootstrap intervals over 669 traces (2,000 resamples) show the probe's AUROC edge over the current-step TF-IDF control at $+0.133$ (CI $[0.111, 0.155]$) and over the position control at $+0.136$ (CI $[0.113, 0.157]$); cross-model transfer cells remain unpaired. A planned counterfactual test would splice the boundary state of an erroneous prefix into a corrected twin and ask whether the model's verdict flips; correction

180 drafts for all 160 pairs exist but 25 await verification, so the run and the causal question it would
181 settle stay open.

182 **Responsible use.** This study uses no human-subject data and deploys no verifier. False alarms,
183 delayed detection, distribution shift, and adversarial traces make the probe unsuitable for high-stakes
184 grading or certification. Consequential calculations still require human or formal verification.

185 Acknowledgments

186 The author thanks the ProcessBench and Qwen teams for releasing the dataset and model check-
187 points used here.

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